

JOÃO SILVA

Data Engineer

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PROFESSIONAL SUMMARY

Results-driven Data Engineer with 2 years of experience building scalable data pipelines and ETL solutions. Proficient in Python, SQL, and cloud technologies with a strong foundation in data warehousing concepts. Passionate about optimizing data workflows and collaborating with analytics teams to deliver actionable insights. Seeking to expand expertise in modern data stack tools and cloud infrastructure.

WORK EXPERIENCE

Data Engineer

DataFlow Systems - Lisbon, Portugal (www.dataflowsystems.io)

March 2023 - Present

Designed and deployed automated ETL pipelines using Apache Airflow to process 500GB+ of daily data from multiple sources including APIs and databases, reducing manual data processing time by 40%.

Built data models and transformations using dbt to standardize metrics across the organization, enabling self-service analytics for 15+ downstream business users.

Developed Python scripts for data validation and quality checks, implementing automated testing frameworks that caught 95% of data inconsistencies before production deployment.

Collaborated with data scientists and analytics teams to understand requirements and translate them into optimized SQL queries and data structures on PostgreSQL and Snowflake.

Junior Data Engineer

Tech Insights Ltd - Lisbon, Portugal

January 2022 - February 2023

Maintained and optimized existing Spark jobs processing structured and semi-structured data, improving query performance by 35% through strategic partitioning and caching strategies.

Built data pipelines using Python and PySpark to ingest data from REST APIs and message queues, storing results in AWS S3 and Redshift for analytics consumption.

Created automated monitoring dashboards using Grafana to track pipeline health, data freshness, and job execution metrics, reducing incident detection time from hours to minutes.

Assisted senior engineers in designing fact and dimension tables following star schema principles, supporting the development of a modern data warehouse serving the analytics team.

Data Analyst Intern

Analytics Hub Co - Lisbon, Portugal (www.analyticshub.co)

June 2021 - December 2021

Wrote complex SQL queries against production databases to extract and analyze customer behavior data, supporting ad-hoc reporting requests from the product and marketing teams.

Automated routine reporting tasks using Python scripts and GitHub Actions, reducing manual report generation overhead by 20 hours per week and improving stakeholder satisfaction.

Performed exploratory data analysis on historical datasets to identify trends and anomalies, presenting findings to non-technical stakeholders with clear visualizations.

Documented data schemas and created technical guides for junior team members, establishing best practices for SQL optimization and data access procedures.

EDUCATION

Bachelor of Science in Computer Science

Instituto Superior Tecnico, Lisbon, Portugal

Graduated: June 2021

TECHNICAL SKILLS

Languages: Python, SQL, Scala, Bash

Data Technologies: Apache Spark, Apache Airflow, dbt, Pandas, NumPy

Databases: PostgreSQL, MySQL, Redshift, Snowflake, MongoDB

Cloud Platforms: AWS (S3, EC2, RDS, Redshift), Azure Data Factory

Tools and Frameworks: Git, Docker, Jupyter Notebook, Linux, Tableau

Concepts: ETL/ELT pipelines, Data Warehousing, Data Modeling, OLAP, Star Schema

CERTIFICATIONS

AWS Certified Cloud Practitioner - Amazon Web Services (2022)

LANGUAGES

Portuguese: Native

English: Fluent

Spanish: Basic

PUBLIC REPOSITORIES

github.com/joaosilva-dev/spark-etl-pipeline - Apache Spark ETL solution for processing CSV and Parquet files

github.com/joaosilva-dev/airflow-dags - Collection of production Airflow DAGs for data pipeline orchestration

github.com/joaosilva-dev/sql-optimization-toolkit - SQL query optimization examples and performance tuning scripts

github.com/joaosilva-dev/python-data-validation - Python framework for automated data quality checks and validations